

GRAMPANCHAYAT MANAGEMENT SYSTEM(GPMS)

Overview

This Django-based Village Management System provides a comprehensive platform for managing various aspects of village administration, including citizen records, land management, tax collection, welfare schemes, and asset tracking. The system serves multiple user roles with distinct functionalities.

Key Features

1. User Authentication System

- Secure login/signup with password hashing
- Session management
- Role-based access control (Admin, Government Monitors, Panchayat Employees, Citizens)

2. Core Modules

Village Dashboard

- Displays village statistics including:
 - Population demographics
 - Financial reports (income/expenditure)
 - Active assets
 - Welfare schemes
 - Recent notifications

Citizen Management

- CRUD operations for citizen records
- Profile management
- Certificate issuance
- Complaint submission

Land Management

- Land ownership tracking
- Agricultural records
- Land transfer history
- Acreage management

Financial System

- Tax calculation and collection
- Payment processing
- Transaction history
- Revenue reports

Welfare Schemes

- Scheme creation and management
- Eligibility checking
- Automated enrollment
- Benefit distribution

Asset Management

- Village asset tracking
- Maintenance records
- Scrap management
- Expenditure tracking

3. User-Specific Functionality

Admin

- Manage government monitors
- Manage panchayat employees
- Activate/deactivate accounts

Government Monitors

- View comprehensive reports
- Analyse village metrics over time
- Monitor scheme effectiveness

Panchayat Employees

- Manage all village records
- Process certificates and complaints
- Update land ownership
- Manage assets and schemes

Citizens

- View personal records
- Pay taxes
- Check scheme eligibility
- View land holdings
- Submit complaints

Technical Implementation

Database

- PostgreSQL backend
- Complex relational schema with 15+ tables
- Proper indexing and constraints

Security

- Password hashing with Django's built-in functions
- Session-based authentication
- Role-based permission checks
- Parameterized queries to prevent SQL injection

Code Structure

- Well-organized views for each functionality
- Comprehensive logging

- Error handling and user feedback
- Database connection management

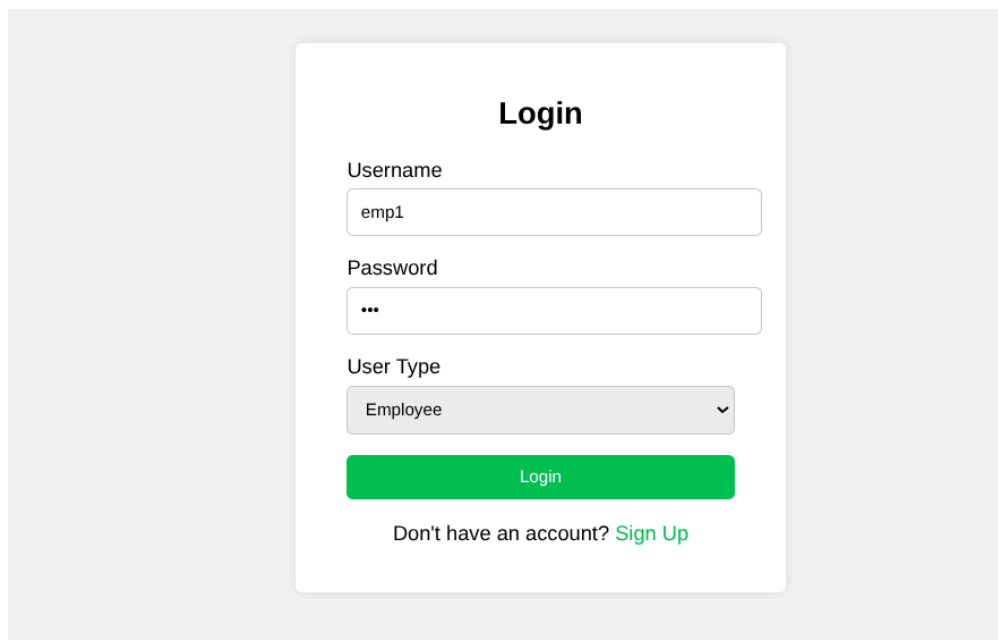
Strengths

1. Comprehensive Coverage: Handles all major aspects of village administration
2. Role-Based Access: Clear separation of concerns between user types
3. Data Integrity: Proper database constraints and transactions
4. User Experience: Informative dashboards and clear navigation
5. Automation: Features like automatic tax calculation and scheme enrollment

Github Link : <https://github.com/adhi0987/GPMS>

Screenshots:

Login-page

A screenshot of a web application's login page. The page has a light gray background. In the center, there is a white rectangular box with rounded corners. Inside this box, the word "Login" is written in bold black text at the top. Below it, there are three input fields: "Username" with the text "emp1", "Password" with three dots, and "User Type" with a dropdown menu showing "Employee". Below these fields is a green rectangular button with the text "Login". At the bottom of the box, there is a link that says "Don't have an account? Sign Up" in green text.

Login

Username
emp1

Password
...

User Type
Employee

Login

Don't have an account? [Sign Up](#)

Employee-Main-page

Panchayat Employee Main Page

Citizens

Lands

Certificates

Taxes

Scheme Enrollment

Welfare Schemes

Assets

Households

View Complaints

Village Dashboard

Logout

Add New Notification

Dynamic-Dashboard



Gram Panchayat Managment System

Home

Login

March 3, 2025:this is five

March 3, 2025:hi

March 3, 2025:This is display of First Notification viewed in notification bar which is displayed in Villagec

Accordion Section

Employees

Assets

Welfare Schemes

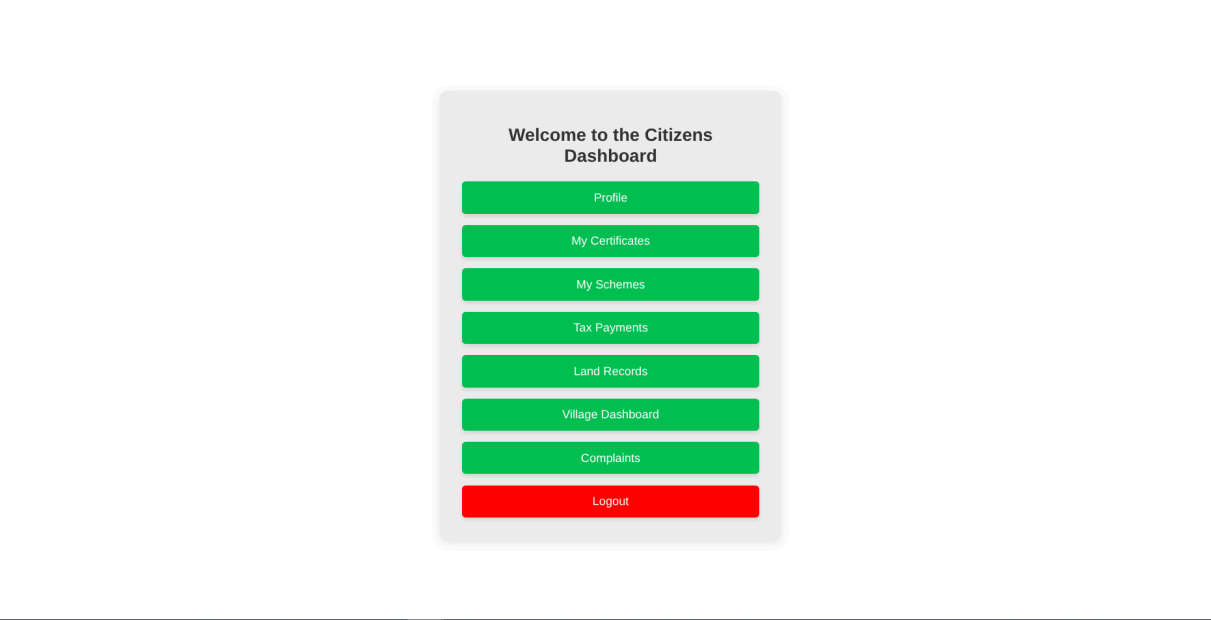
Population

Total Income To Panchayat

Total Expenditure To Panchayat

Back

Citizen-Dashboard



Government-Monitor-Main-Page

